Instructions to Lab 1 Instructions on Simulation on Transmission Line Using CST Studio Suite

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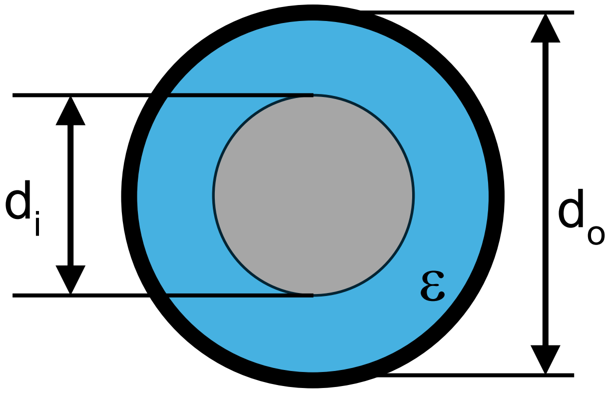
Aim

To simulate parameters of a transmission line and check the matchiness to the load.

Background

The transmission line we are going to simulate is a coaxial cable, with characteristic impedance of $Z_0 = \frac{60}{\sqrt{\varepsilon_r}} \ln\left(\frac{d_o}{d_i}\right)$, where ε_r is the relative permittivity of the insulator between the conductors, d_o is inner diameter of the outer conductor, d_i is the diameter of the inner conductor, as shown in the figure below.

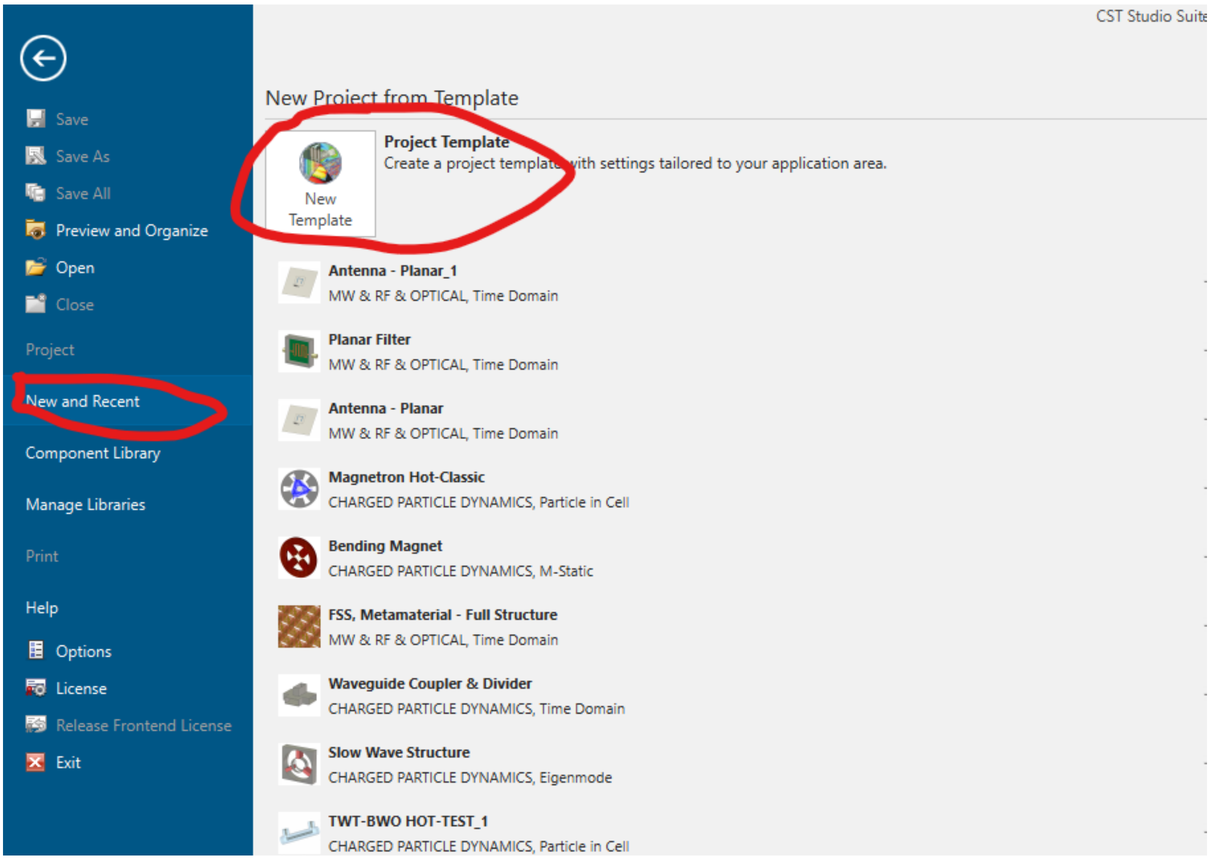


Procedures

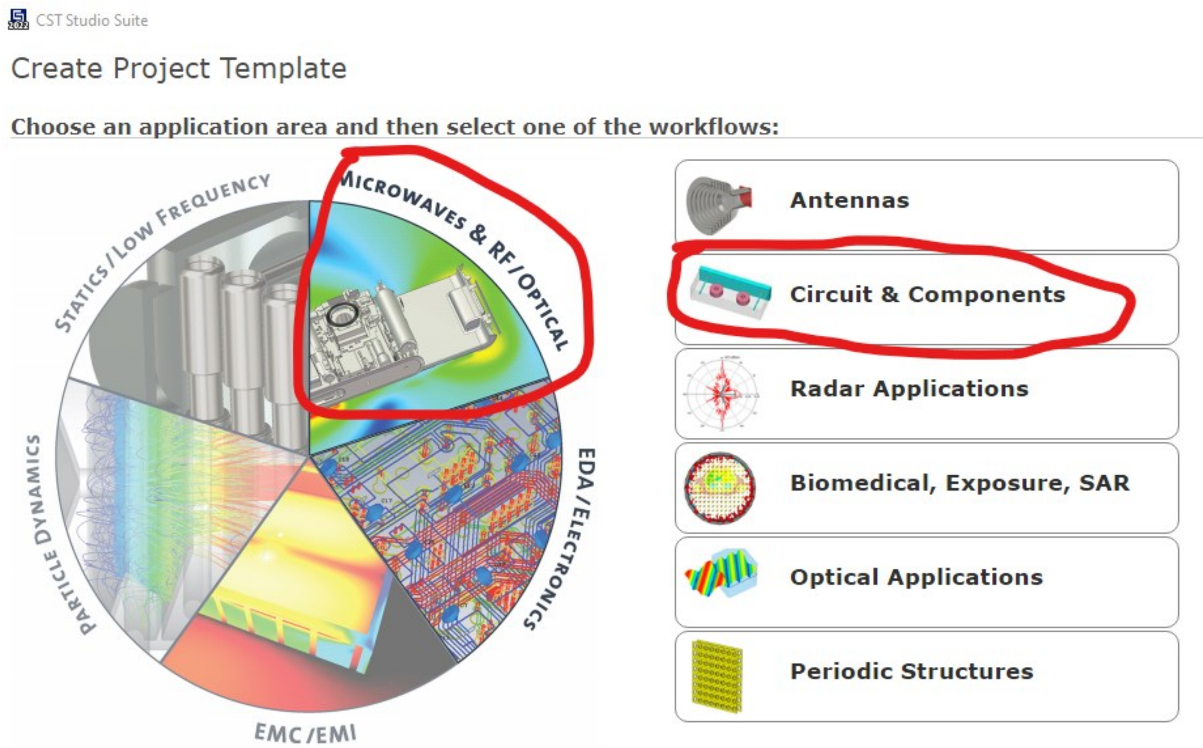
The procedures of simulation in CST Studio Suite are as follows.

1. **Open CST and Set a template.**

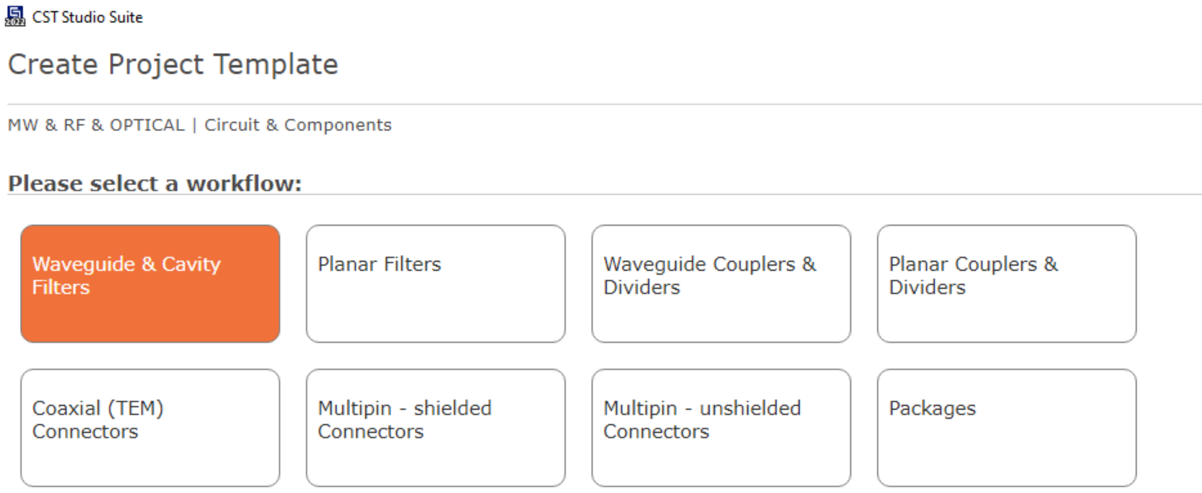
Open CST Studio Suite, → New and Recent → New Template



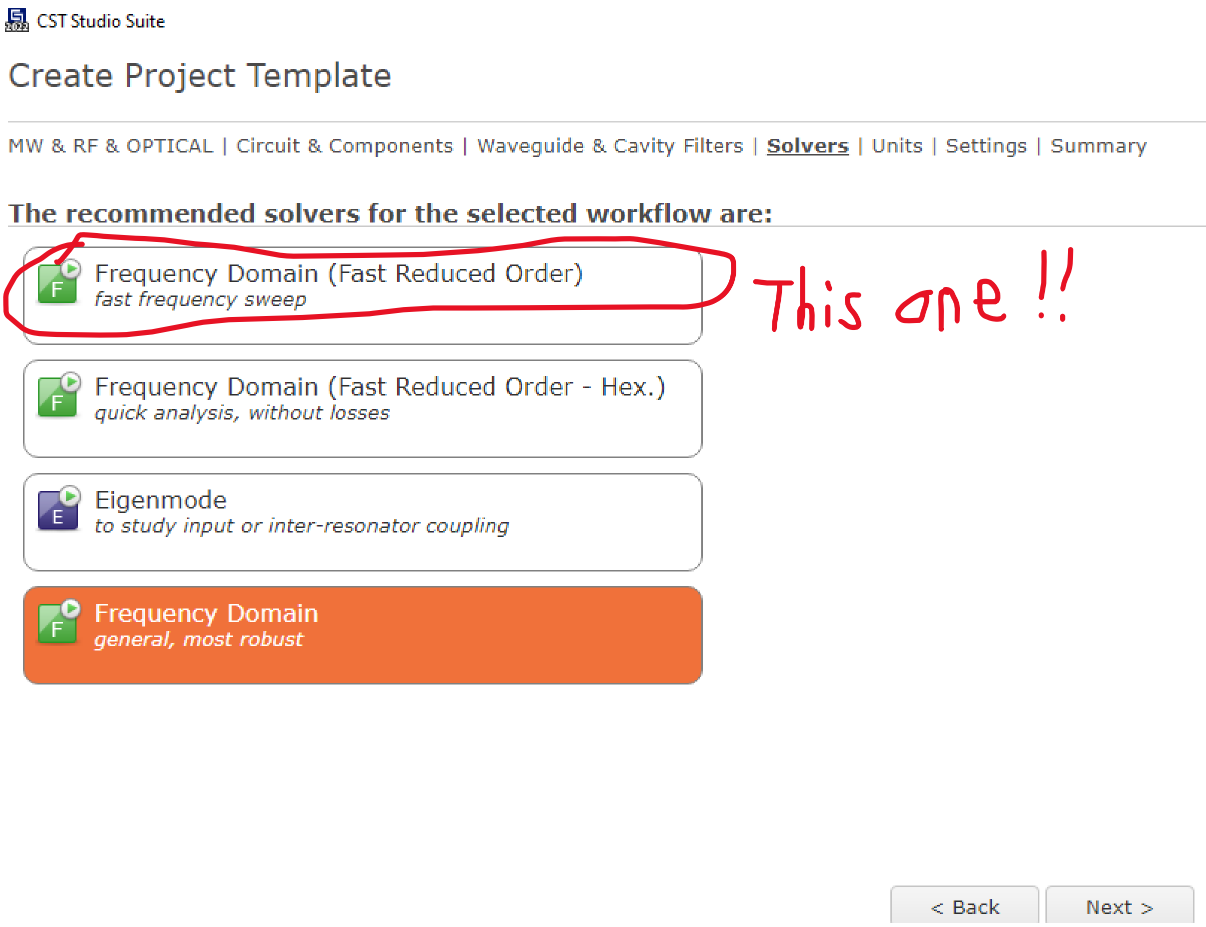
Then, choose Microwaves & RF/Optical→ Circuit & Components



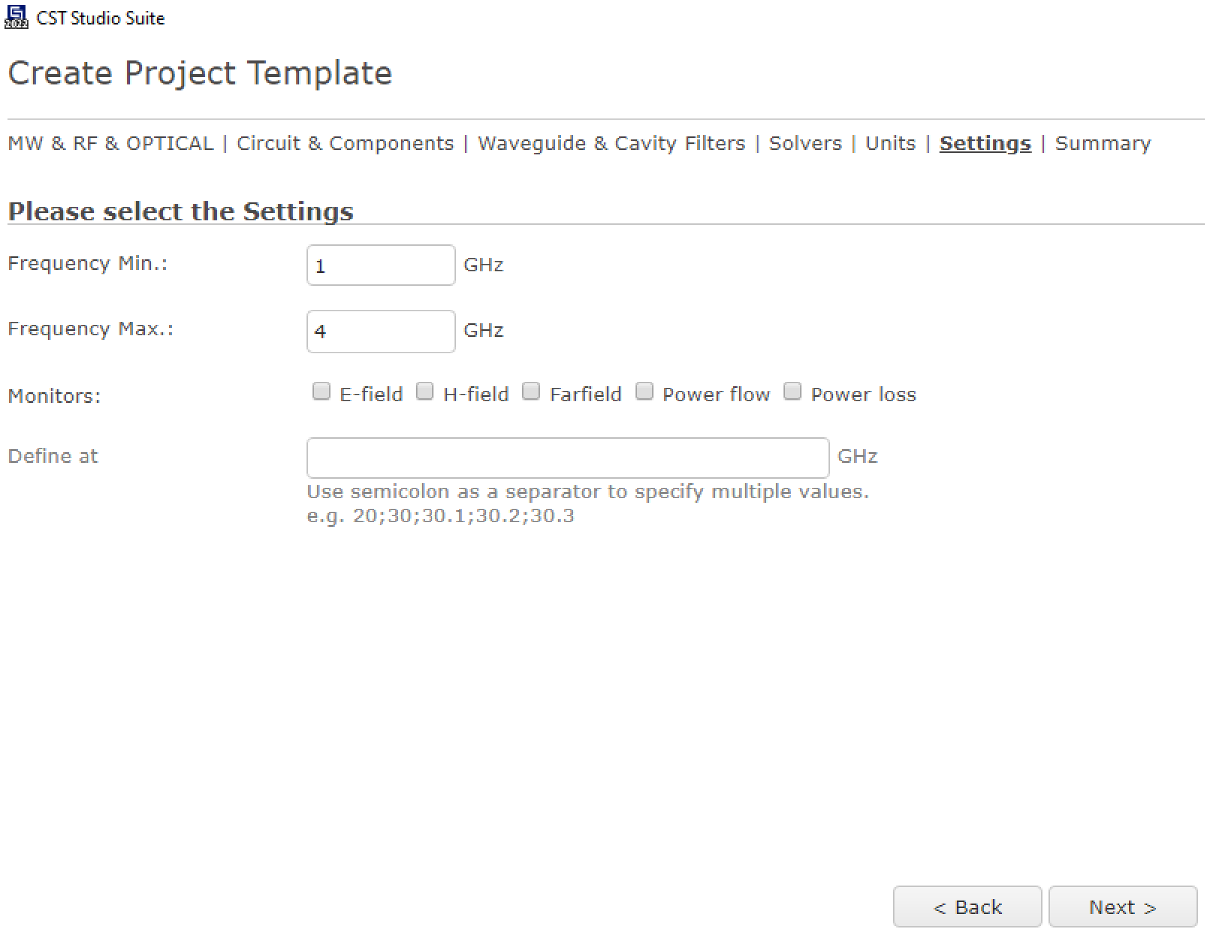
Then choose Waveguide & Cavity Filters, then ‘Next’



Then choose, ‘Frequency Domain (fast reduced order)’



Then ‘Next’

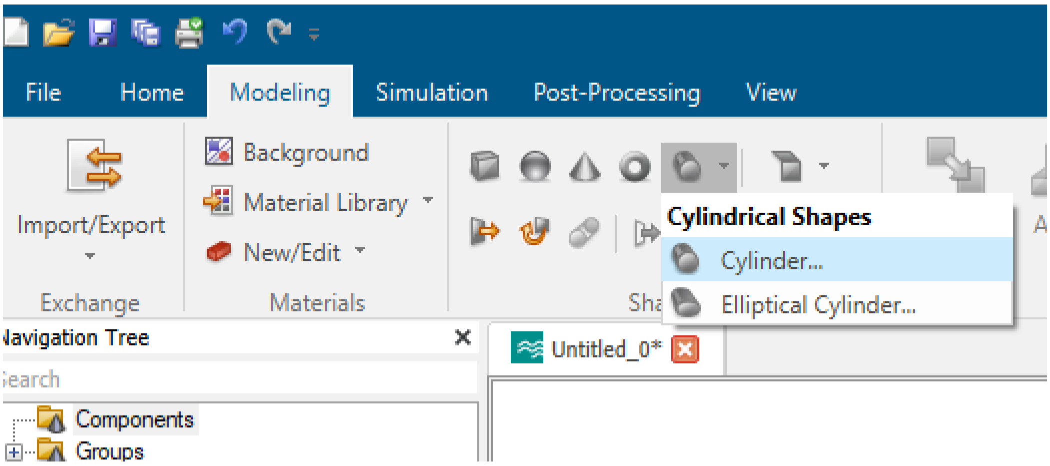


Then set frequency range: 1-4 GHz. Then 'Finish'.

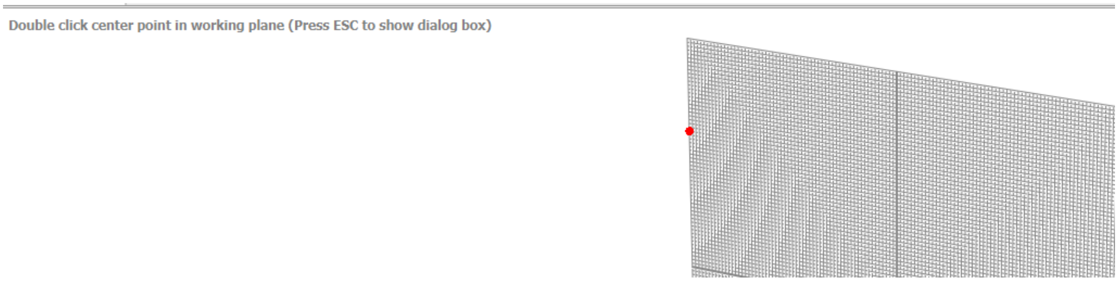
2. Building geometrical structures

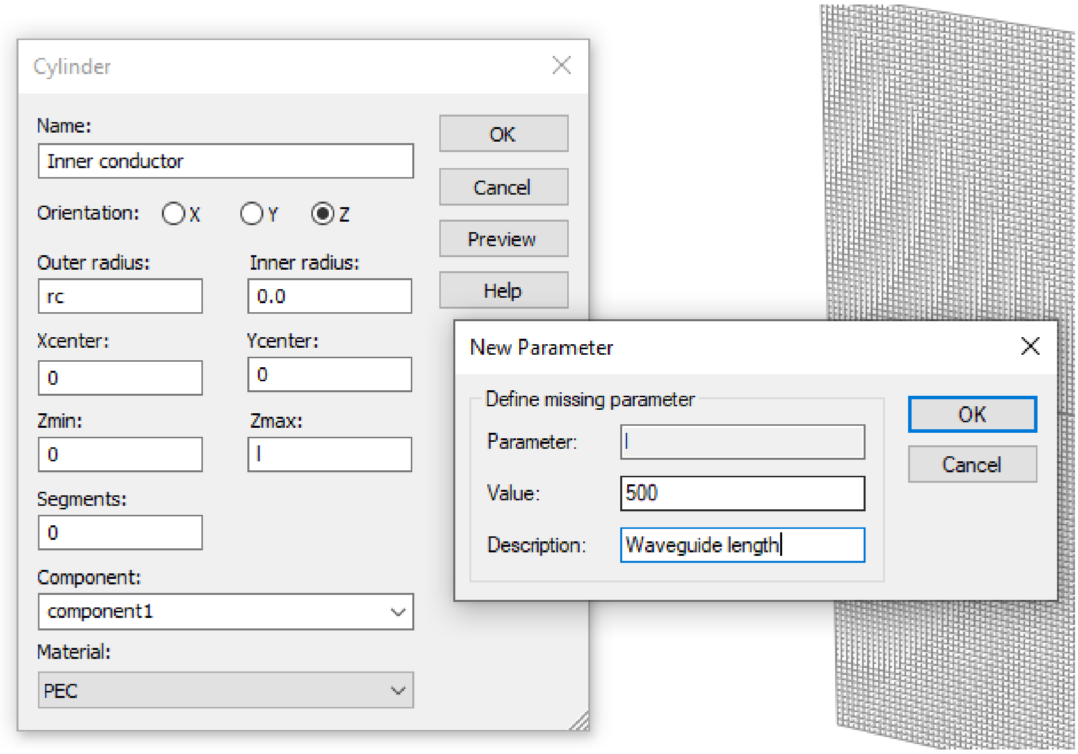
2.1 Build inner conductor.

Choose 'Modeling' → Cylindrical Shapes → Cylinder.



Then press ESC (on your keyboard!!!) to show dialog box, as shown below.

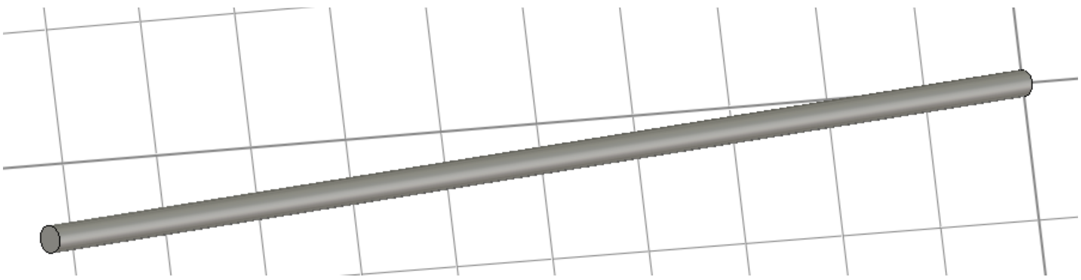




Set the inner conductor parameters. You can either:

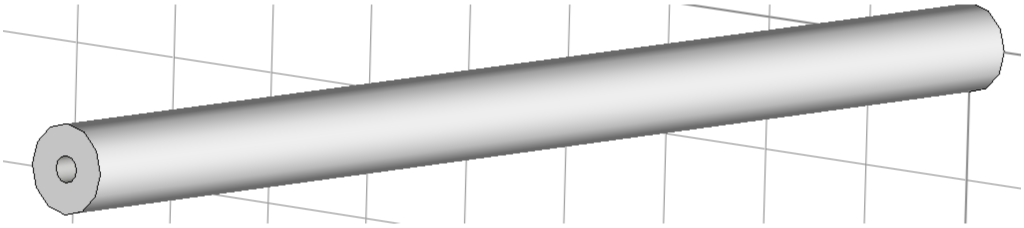
- Use parameters such as *rc* and *l* , for inner conductor radius and length. Set them as 5 mm and 500 nm;
- Or directly type in 5, and 500 as outer radius and Z_{max}.

You'll get a cylinder as follows. Using PEC for material.



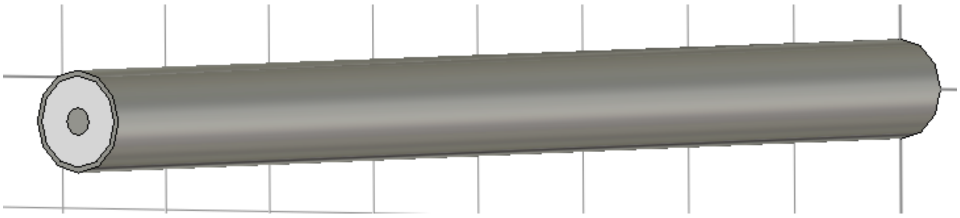
2.2 Building Teflon insulation

Similarly build insulation by creating cylinder with inner radius (rc, the same as the previous step), outer radius (rt=33.5/2), and the same length l=500 mm. Creating a new material for Teflon with ε_r=2.1, and use this for the insulation.



2.3 Building Outer Conductor

Similarly build outer conductor by creating cylinder with inner radius (rt=33.5/2), outer radius (rt=33.5/2+1), and the same length l=500 mm, using PEC as material.

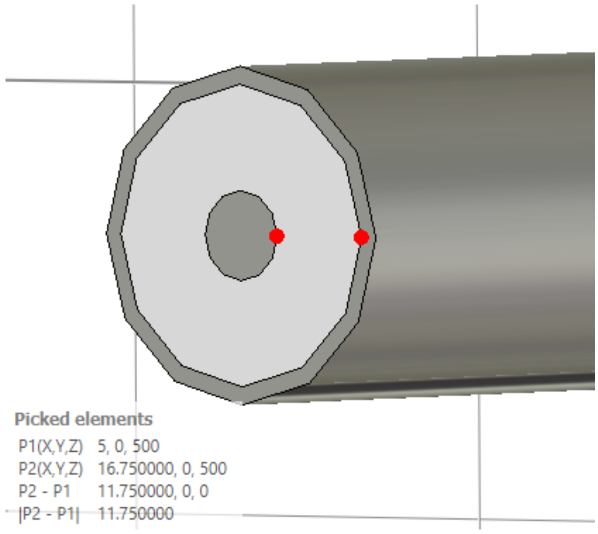


3. Setting 2 ports: port 1 and port 2 for source (generator) and load impedance.

Modelling→ Pick point→ Pick point on circle: Choose a point on outer edge of inner conductor.

Pick point→ Pick point on circle: Choose a point on inner edge of outer conductor.

You have got 2 points.



Then Simulation → Discrete Port, Choose S-parameters, Set impedance as 50 Ohm. Click OK. Now you have defined a port.

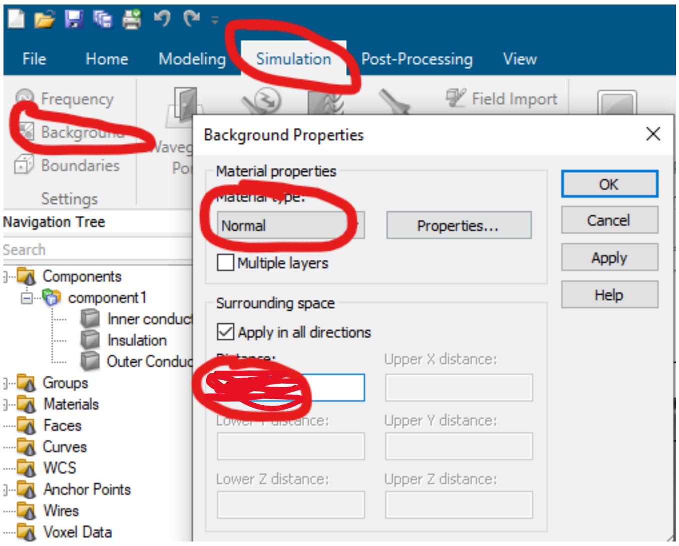
Rotate to the other end of the co-axial cable, define another port using the same way.

S parameters mean how much is transferred from one terminal to the other.

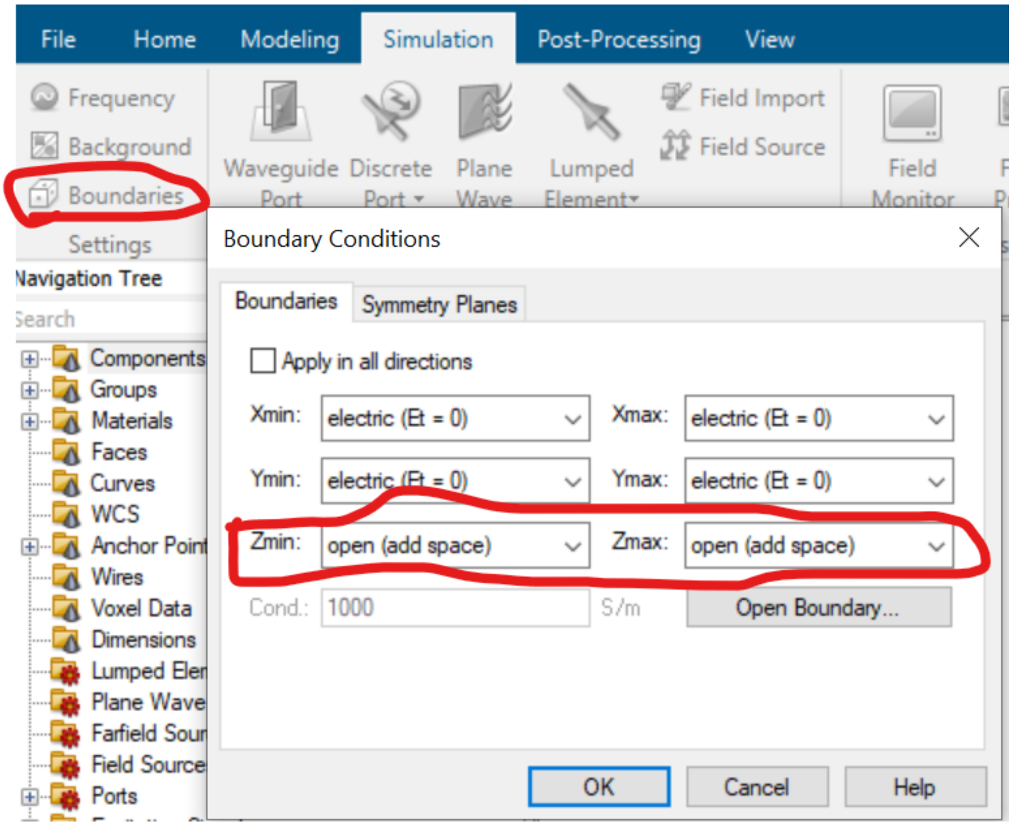
4. Simulation settings.

Check Simulation→Frequency, f_{min}= 1 GHz, f_{max} = 4 GHz.

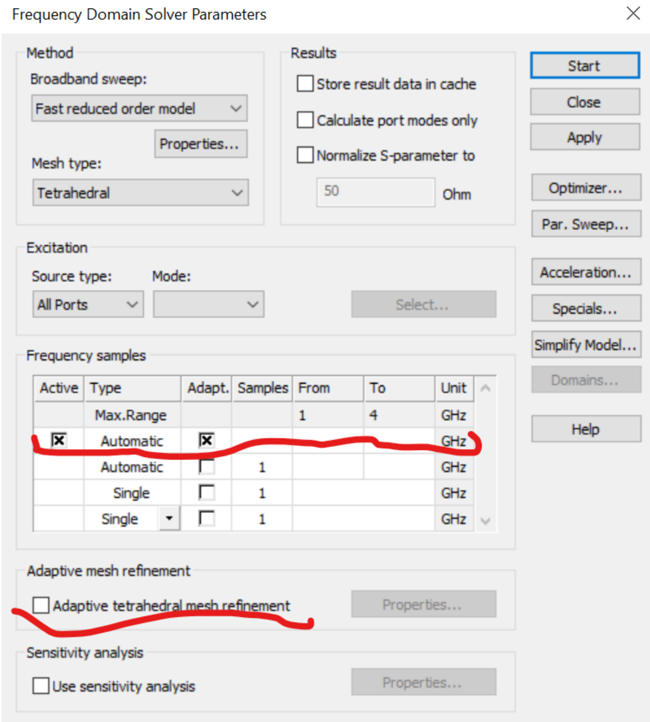
Simulation→Background→Normal.



Boundaries→ Set Z_{min}, Z_{max} as 'Open (add space)'.



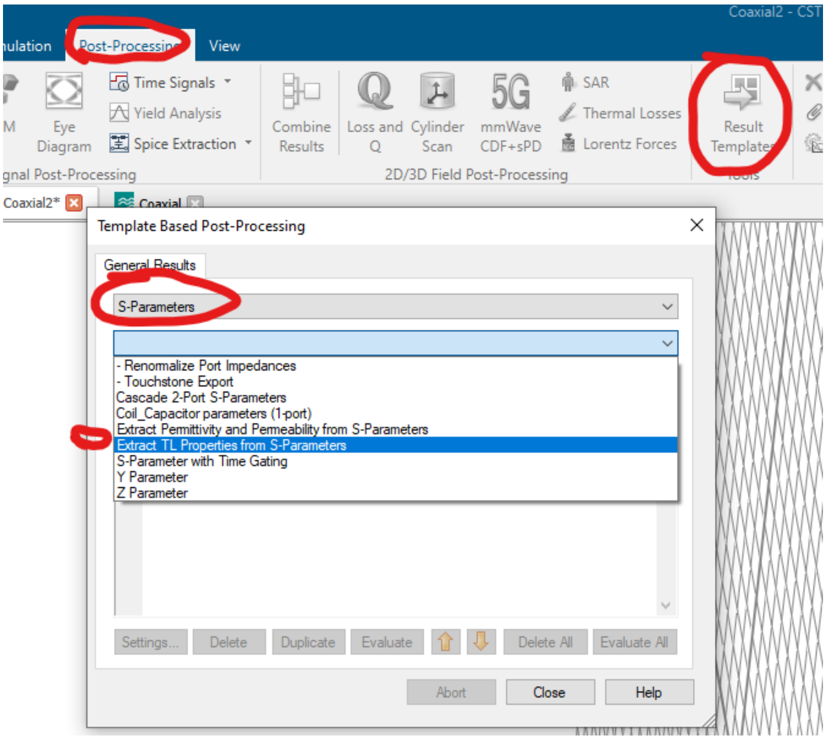
5. Running simulation
   Simulation→F solver, and make the settings in the solver window.



Click on 'Start' to start simulation.

6. Post processing.

After simulation, set the post-processing: results templates→ S parameters → Extract TL properties → Evaluate.



Check the results here.